A12749W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMICS

TRINITY TERM 2018

Saturday 26th May 2018, 14:30 - 17:30

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

Answer **all** *questions in Part A and* **two** *long questions from Part B.*
Part A attracts 1/3 of the marks in total; each question in Part B attracts 1/3 of the marks.

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets;
in Part B, the weight assigned to each part of the multipart question is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

Part A

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets

1. Consider an economy in which each firm can produce output X using labour input L according to a production function $X = AL^{\alpha}/\alpha$, where A is a technology parameter and $0 < \alpha < 1$. Suppose that the price of X is p, the price of L is w, and production is chosen by each firm to maximise profits.

   (a) Derive a firm's labour demand function, output supply function, and its level of profits as functions of p and w.

   (b) What proportion of the value of output goes as profits, and what proportion as wages?

   There are M capitalists, each of whom owns a single firm, and N workers, each endowed with a single unit of labour. All the capitalists and workers spend their entire income (from profits and wages respectively) on output, and their utility is equal to units of X consumed. Henceforward, take one unit of labour as the numeraire.

   (c) Show that the equilibrium value of p is given by $(pA)^{1/(1-\alpha)} = N/M$.

   (d) Technical progress improves the efficiency of the technology.
   What are the equilibrium effects of such an increase in the parameter A?
   How are the benefits of the increase distributed between capitalists and workers?

[30%]

2. What is the definition of a Nash equilibrium of a game?
   Does the existence of a Nash equilibrium lead to a good prediction of how the players will play a game?

   Consider the two-player, one-shot, simultaneous-move game, which depends on the parameter t:

	Left	*Right*
Up	t 0	3 3
Down	0 5	5 0

   where the row player receives the payoff in the bottom-left corner of the cells, and the column player receives the payoff in the top-right corner of the cells.

   (a) Find all the Nash equilibria when $t < 3$, say $t = 2$.

   (b) Find all the Nash equilibria when $t > 3$, say $t = 4$.

   (c) What if $t = 3$?

[20%]

3. Janet is an expected utility maximiser whose utility of wealth is $u(w) = \ln w$. Her initial wealth is 64, and she can invest in a risky project with a net return of +26 or −24, each equally likely. (So, if it succeeds, her final wealth will be 90, but if it fails, her final wealth will be 40.)

   (a) What is the expected net return of the project?
   What is Janet's risk premium for the project?
   Show that she won't undertake the project by herself.

   She asks Sam (who has the same preferences and initial wealth as Janet) to share the risky project with her – they would each get a net return of either +13 or −12.

   (b) What is their individual expected net return and their risk premium for their share of the project?
   Will the two of them accept the opportunity to share the project?
   If not, how many similar people do they need?
   Explain why sharing with more people makes the project more acceptable.

   (c) What size of group would maximise the expected utility of each member?
   Why does sharing become less beneficial when the group gets bigger than this?

[25%]

4. Three qualities of second-hand bicycles are available in equal numbers. There are many buyers and sellers; they have different values for each quality of bicycle, and everyone knows the value that each type of consumer attaches to each quality of bicycle.

Quality	Buyer's value	Seller's value
High	100	75
Medium	65	60
Low	30	45

   (a) What is the efficient outcome?

   Buyers do not know the quality of any particular bicycle for sale, but sellers do know the quality of the one that they own. The price at which any bicycle is traded will be determined by supply and demand. No-one wants more than one bicycle.

   (b) What is the equilibrium outcome?

   Now assume that the sellers can offer a reimbursement of 50 to the buyer, payable if the bicycle breaks down; those of high quality never fail, those of low quality always fail, and those of medium quality fail with probability $\frac{1}{2}$.

   (c) Show that there is an equilibrium with $85 \leq p < 95$ that is efficient.

   (d) Are there any other (inefficient) equilibria in which trade occurs?

[25%]

Part B

5. "Taxes cause economic inefficiency."
   (a) Explain the reasoning underpinning this statement.
   (b) Give examples of two different types of tax for which this statement need not be true, explaining in detail why not.
   (c) What are the implications of your analysis for reform of the tax system?

6. What techniques are available to estimate the expected social value of a proposed public investment project (e.g. an infrastructure project)?

   Why might the techniques and valuations used for a public project differ from those used in calculating a private commercial rate of return?

7. "A merger between two rival firms is a form of collusion. Anti-trust authorities should ban such mergers, just as they penalize firms who collude in other ways."
   Discuss.

8. What is risk aversion? What do we mean when we say that Greg is more risk averse than Jamal? How might we test whether or not that assertion is true?

   What does "lottery A is more risky than lottery B" mean? How might we show this in theory, and in practice?

9. There is a project with an uncertain outcome (high or low) that depends on the level of effort (high or low) exerted by whoever undertakes it – the high outcome is more likely when effort is high than when effort is low. If a social planner were making the decision, she would want high effort to be exerted. However, there is a risk-neutral principal that owns the project, and an agent who will undertake it on their behalf.

   If the agent is also risk neutral, explain what the principal would do and why. What level of effort will be exerted?

   From now on, assume that the agent is risk averse.

   When effort is observable, what level of effort will the principal specify, and how will they determine the wage?

   When effort is not observable, will the principal want to specify the same level of effort?

10. Alex's has a house worth h and some financial assets worth f. Alex's house is in an area that is vulnerable to hurricanes; if there is a violent storm then the house will flood and be worth nothing, but the financial assets will be unaffected; if there is no flood the value of all her wealth will be unchanged. The probability of flooding is π.

Insurance against flood damage is available at a premium of p per unit of cover, so an amount of cover q costs $p\,q$.
However, $p > \pi$.

(a) In a diagram illustrating wealth in the two states 'no flood', 'flood', mark the points indicating her final wealth if she buys no insurance, and if she buys full insurance.

If Alex buys an amount of cover q (with $0 \leq q \leq h$), what is her final wealth in the two states 'no flood', 'flood'?
Mark such a point on your diagram. [15%]

(b) Alex is an expected utility maximiser, and has a utility function $u(\cdot)$ that is increasing and concave.

Does this mean that she is risk averse?

Write down her expected utility $\mathrm{E}[u]$ if she buys an amount of cover q.
By examining the value of $\partial \mathrm{E}[u]/\partial q$ when $q = h$, or otherwise, explain why Alex will not choose to fully insure the house. [30%]

(c) Let $u(y) = \ln(y)$.
Assuming that Alex wants to buy *some* insurance, find q^*, the optimal amount of cover.

What condition, in terms of h, f, π and p, would imply that $q^* = 0$?
How does that relate to the slope of Alex's indifference curve at the endowment? [30%]

(d) When $q^* > 0$, how would it change if

(i) the value of Alex's house increased,

(ii) the value of Alex's financial assets increased?

Comment on each of these changes. [25%]